

In a data stream, data continuously comes in one after another. Thus, you do not see all the data at the beginning. You need to write an efficient python program to find median in a data stream. Assume that the all the elements are integers.

Submission guidelines:

Submit this as <midsem-roll>.py file. The data file will be supplied from the command line.

Output format: You need to print the median after every element you read from the file.

For example, if the file contains: 5, 2, 7, 9, 1, then you should print 5, 3.5, 5, 6, 5.

Please note that the median of a set of elements with even number of elements is the average of the two middle elements.

Grading policy: Total Marks: 30

- No marks, if you sort every time you see a new element.
 - No marks if you sort the entire array and then find median. This violates the stream data property.
 - no marks, if the time taken is $O(n^2)$ or more.
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- **Deadline: 4.30 pm today (14.09.2026)**